

DEMO — ILLUSTRATIVE DATA ONLY

SITE INTELLIGENCE REPORT

11822 E 116th St N

Collinsville, OK 74021

36.32197°N, 95.84203°W | Tulsa County



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Executive Summary

Site Identification

ADDRESS	11822 E 116th St N Collinsville, OK 74021
COUNTY	Tulsa County
COORDINATES	36.32197°N, 95.84203°W
IECC ZONE	Zone 3 – Mixed-Humid
POP. (5 MI)	63,009 (est.)

Key Observations

FEMA Flood Zone X – area of minimal flood hazard per NFHL mapping.

Seismic PGA 0.078g – Low seismic hazard. See Section 2 for details.

Wildfire Hazard Potential: Very Low (46 / 144,153 national index).

Risk & Context Summary

Tabular data summary. Values presented as reported by source datasets without modification. No risk rating, scoring, or suitability determination is made by this report.

Category	Reported Value	Source
Flood Risk	Zone X – Minimal	FEMA NFHL
Seismic PGA	0.078g (MCE _r)	USGS NEHRP 2020
Wildfire WHP	46 / 144,153	USDA Forest Service 2023
Hydrologic Group	Unavailable – developed/urban area	USDA SSURGO
FEMA NRI Score	45.2 – Relatively Low	FEMA National Risk Index

Environmental Risk

Flood Risk – FEMA National Flood Hazard Layer

FLOOD ZONE	Zone X – AREA OF MINIMAL FLOOD HAZARD
SFHA (100-YR)	No – not within Special Flood Hazard Area per NFHL
BASE FLOOD ELEV.	Not assigned for Zone X
NFHL DATE	2026-03-09
MAPPED AREA	Yes – FEMA flood mapping available for this area

FEMA flood zone status reflects the NFHL map effective 2026-03-09. Map amendments (LOMA/LOMR), remapping projects, or local ordinance changes may affect current status. Verify current FIRM panel and insurance requirements with your lender, local floodplain administrator, and a licensed flood specialist before relying on this data. Approximately 15–20% of the US has not been mapped by FEMA – unmapped areas are explicitly noted in this report and must never be assumed to be Zone X. See FIRMette (<https://msc.fema.gov>) for the authoritative current map.

Seismic Risk – USGS NEHRP 2020 Seismic Hazard Maps

PGA (MCE _r , 2%/50YR)	0.078g – Low
S ₀ (0.2S SPECTRAL)	0.17g
S ₁ (1.0S SPECTRAL)	0.08g
HISTORICAL EQ ACTIVITY	4 events ≥M3.0 within 50 mi, last 50 years. Max magnitude: M3.7.

About Peak Ground Acceleration (PGA)

PGA is the maximum expected ground acceleration at a site, expressed as a fraction of gravitational acceleration (g). It reflects a 2% probability of exceedance in 50 years (the MCE_r level used in building codes). National range and general hazard tiers:

PGA Range	General Hazard Level	Typical Regions
< 0.05g	Very Low	Central & eastern plains, much of the South
0.05 – 0.15g	Low	Midwest, mid-Atlantic, parts of the Southeast
0.15 – 0.30g	Moderate	Central US, Intermountain West
0.30 – 0.60g	High	Pacific Northwest, parts of Utah/Nevada
> 0.60g	Very High	Coastal California, Cascadia subduction zone

These tiers are general context only and do not constitute a seismic design classification. Seismic Design Category (SDC) per ASCE 7 requires a licensed structural engineer. For interactive maps and full methodology, see <https://earthquake.usgs.gov/hazards/>.

Seismic Design Category (SDC) is an engineering determination per ASCE 7 and requires a licensed structural engineer. Hazard values are probabilistic estimates from the USGS national hazard model and do not constitute a site-specific geotechnical evaluation. Source: [USGS NEHRP 2020 Seismic Hazard Maps](#).

Wildfire Hazard Potential – USDA Forest Service

WHP INDEX VALUE **46 – Very Low** (national max: 144153)

DATA VINTAGE 2023 (270m resolution)

About Wildfire Hazard Potential (WHP)

WHP is a continuous index produced by the USDA Forest Service Fire Modeling Institute. It reflects the relative potential for wildfire that would be difficult to control, based on fire weather, fuel conditions, and topography. It is *not* a probability of fire occurrence. The continuous index is typically grouped into five general classes:

WHP Range (approx.)	Class
1 - 2,249	Very Low
2,250 - 8,999	Low
9,000 - 35,999	Moderate
36,000 - 143,999	High
≥ 144,000	Very High

Class breakpoints are approximate and may vary by dataset edition. For full methodology and downloads, see <https://www.firelab.org/project/wildfire-hazard-potential>.

WHP is a modeled index, not a regulatory classification. Wildland-Urban Interface (WUI) status and local fire hazard severity zones (FHSZ) may impose additional construction and insurance requirements. Consult your local fire authority and AHJ. Source: [USDA Forest Service Fire Modeling Institute, 2023](#).

FEMA National Risk Index – Composite Hazard Risk Score

CENSUS TRACT 40143005811

COMPOSITE FEMA NRI SCORE 45.2 – Relatively Low

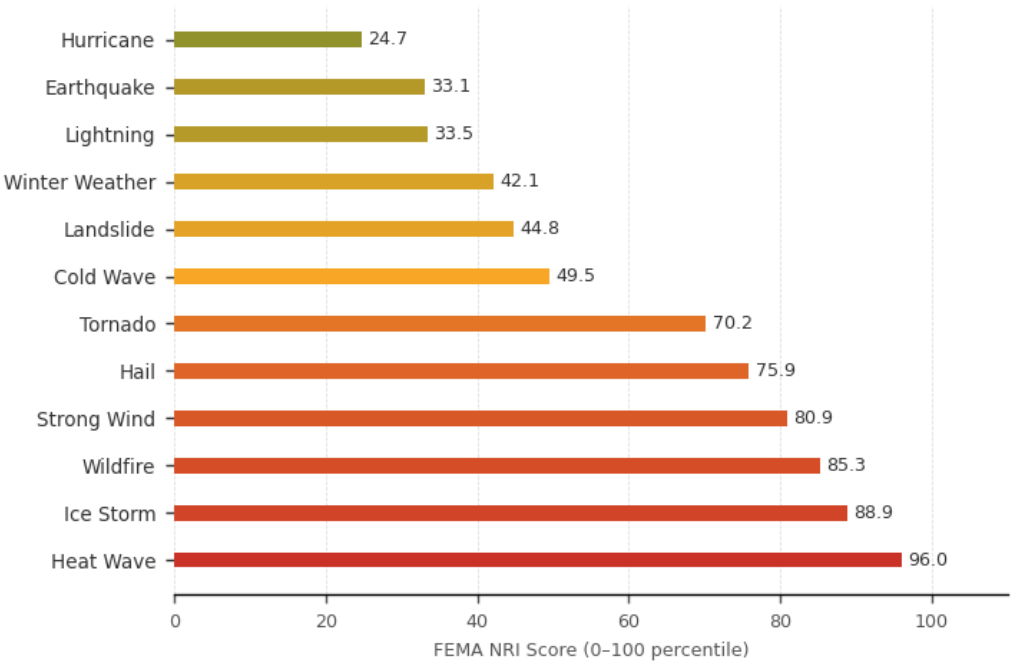
About the FEMA National Risk Index Score

The NRI composite score is a percentile-based value (0-100) representing a census tract's relative natural hazard risk compared to all other US tracts, incorporating expected annual loss, social vulnerability, and community resilience. FEMA uses five rating categories:

Score Range	FEMA Rating
0 - 20	Very Low
20 - 40	Relatively Low
40 - 60	Relatively Moderate
60 - 80	Relatively High
80 - 100	Very High

Component hazard scores use the same 0-100 percentile scale relative to other US tracts. A higher score indicates higher relative risk for that hazard compared to the national distribution. For full methodology, data dictionary, and interactive map, see <https://hazards.fema.gov/nri/>.

FEMA National Risk Index – Component Hazard Risk Scores



Physical Context

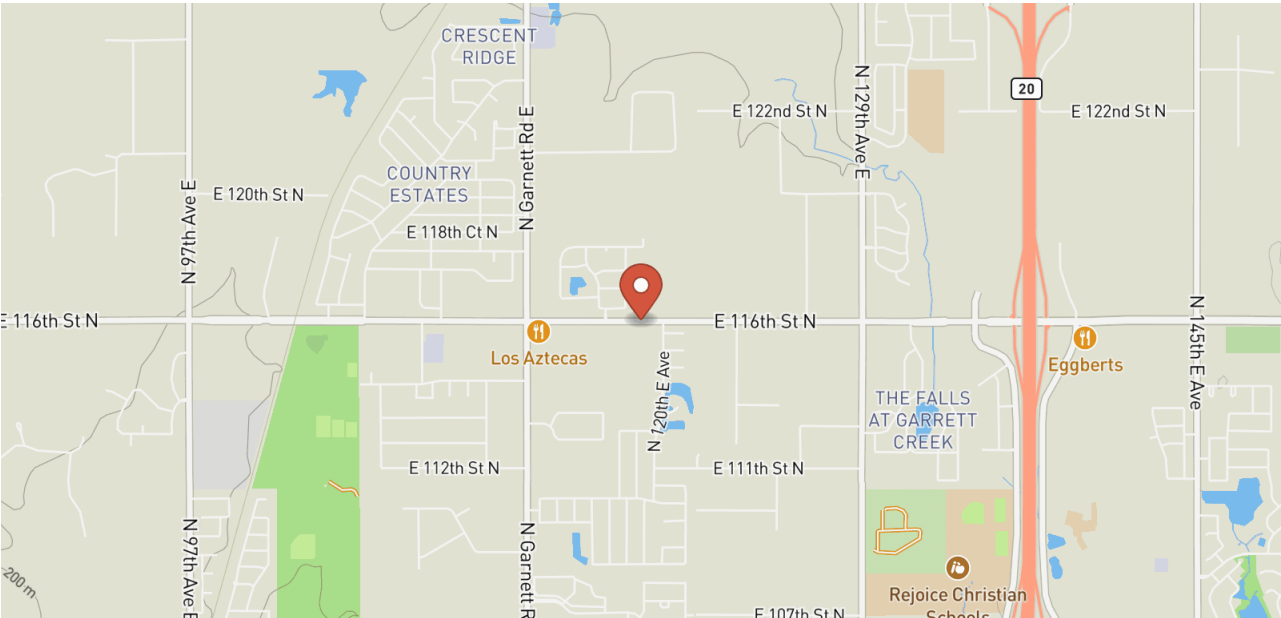
Elevation – USGS 3DEP (10m)

SITE ELEVATION	687.8 ft (209.6 m)
DEM RESOLUTION	25.3m (1/3 arc-second)

Slope – USGS 3DEP Derived

MEAN SLOPE	1.5%
MAX SLOPE	2.3%
RADIUS	100.0m
REFERENCE BANDS	0-5% / 5-15% / >15%

Slope values are DEM-derived estimates (100.0m radius). Site-specific grading conditions should be confirmed by a licensed surveyor or civil engineer.



Land Cover – USGS NLCD 2021 (500.0m radius)

DOMINANT CLASS **Pasture/Hay**
NLCD CODE 81
IMPERVIOUS SURFACE 57.1% (estimated)

Land Cover Class	% of Radius
Pasture/Hay	38.1%
Developed, Low Intensity	21.6%
Developed, Open Space	17.1%
Developed, Medium Intensity	17.1%
Deciduous Forest	2.9%
Developed, High Intensity	1.2%
Open Water	0.7%
Mixed Forest	0.6%
Grassland/Herbaceous	0.5%
Woody Wetlands	0.2%

Hydrography – NHDPlus High Resolution

NEAREST WATERBODY Unnamed – 2.14 mi
NEAREST FLOWLINE Unnamed – 0.15 mi
SEARCH RADIUS 5 miles

Climate & Weather

CLIMATE STATION	TULSA INTL AP, OK US (USW00013968)
DISTANCE FROM SITE	8.8 miles
NORMALS PERIOD	NOAA 1991-2020 Climate Normals
IECC ZONE	Zone 3 – Mixed-Humid. Local jurisdiction adoption may vary. Verify with AHJ.

3451 HDD ANNUAL	2128 CDD ANNUAL	73% AVG REL. HUMIDITY	90 FREEZE-THAW DAYS
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Monthly Climate Normals (1991-2020)

Metric	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Max Temp (°F)	49	54	63	72	80	88	94	93	85	74	61	51
Min Temp (°F)	28	32	41	50	60	69	73	72	63	51	39	31
Precip (in)	1.63	1.62	3.10	4.37	5.73	4.65	3.76	3.38	3.85	3.78	2.66	2.43
Snowfall (in)	1.9	2.4	1.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.8	1.7

Metric Definitions

Heating Degree Days (HDD)

The sum of degrees per day that the mean temperature falls below 65°F over a year. A day averaging 45°F contributes 20 HDD. Used to estimate heating energy demand – higher HDD = more heating needed.

Cooling Degree Days (CDD)

The sum of degrees per day that the mean temperature exceeds 65°F over a year. A day averaging 85°F contributes 20 CDD. Used to estimate cooling/AC energy demand.

Freeze-Thaw Days

The number of days per year where the temperature crosses the 32°F (0°C) threshold – either rising above or dropping below freezing within the same day. Relevant for pavement and concrete durability, since repeated freeze-thaw cycles cause cracking and spalling.

Climate values are NOAA 1991-2020 climate normals measured at TULSA INTL AP, OK US, 8.8 miles from the site. Station-based normals may not reflect site-level microclimatic variation due to local topography, urban heat island effects, or distance from station. Source: NOAA National Centers for Environmental Information.

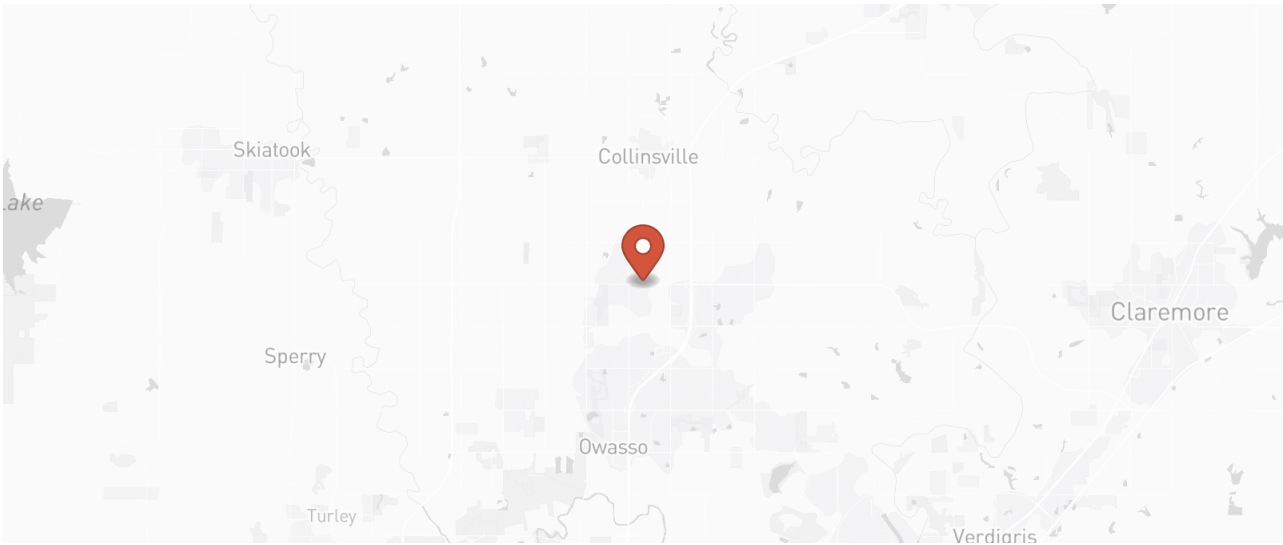
Solar Resource

ANNUAL AVG GHI 4.4803 kWh/m²/day peak sun hours

LATITUDE 36.32197414685°N

Monthly Solar Resource (GHI, kWh/m²/day)

Metric	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
GHI	2.53	3.15	4.16	5.19	5.78	6.70	6.63	5.94	4.98	3.71	2.74	2.18
Sunrise		1:10	12:33	11:49	11:16	11:05	11:17	11:41	12:06	12:30	1:01	1:27
Sunset		12:03	12:30	12:56	1:22	1:42	1:41	1:14	12:31	11:46	11:14	11:09



Solar resource values are long-term climatological averages (NASA POWER, 2001–2020) at the grid cell containing this location (≈0.5° resolution). They represent horizontal surface irradiance and do not account for shading, panel tilt, azimuth, or system losses. No energy production estimate is made or implied. Source: [NASA POWER \(https://power.larc.nasa.gov/\)](https://power.larc.nasa.gov/), accessed March 9, 2026.

Demographic & Market Context

Area-weighted block group intersections. Straight-line radius buffers – physical barriers not accounted for. Source: 2019-2023 ACS 5-Year Estimates.

Variable	1-Mile Radius	5-Mile Radius	10-Mile Radius
POPULATION & AGE			
Total Population	5,145	63,009	104,441
Median Age	32.1	36.6	37.5
Average Household Size	3.08	2.74	2.76
HOUSEHOLDS & INCOME			
Total Households	1,706	23,374	38,399
Median Household Income	\$98,395	\$88,834	\$86,095
Per Capita Income	\$36,486	\$38,673	\$38,180
% Below Poverty Level	N/A	N/A	N/A
% Receiving Public Assistance	0.6%	1.4%	1.9%
HOUSING STOCK			
Total Housing Units	1,827	24,650	41,507
Median Home Value	\$208,227	\$244,805	\$229,293
Median Gross Rent	\$1,497	\$1,264	\$1,085
Vacancy Rate	6.6%	5.2%	7.5%
Median Year Built	2002	1997	1993
TENURE			
% Owner-Occupied	81.1%	72.4%	74.6%
% Renter-Occupied	18.9%	27.6%	25.4%
EDUCATION			
% Bachelor's Degree or Higher	23.7%	32.0%	29.0%
% High School Diploma or Higher	94.5%	94.1%	91.9%
COMMUTE & EMPLOYMENT			
Mean Travel Time to Work (min)	N/A	N/A	N/A
% Drive Alone	78.4%	80.8%	80.3%
% Work from Home	9.7%	10.5%	10.2%
Unemployment Rate	4.0%	4.4%	4.8%

Variable	1-Mile Radius	5-Mile Radius	10-Mile Radius
Labor Force Participation Rate	74.5%	69.7%	66.8%
RACE & ETHNICITY			
% Non-Hispanic White	68.4%	70.5%	66.8%
% Hispanic or Latino	6.1%	7.5%	8.5%
% Black or African American	1.8%	2.7%	5.1%
% Asian	0.2%	2.5%	2.5%
% Other / Multiracial	23.5%	16.8%	17.0%
LANGUAGE			
% Limited English Proficiency	N/A	N/A	N/A

Demographic values are area-weighted estimates derived from Census block groups intersecting straight-line radius buffers. Physical barriers (rivers, highways, etc.) are not accounted for. Population counts may differ from administrative area totals. Radius buffers use geodesic measurement. All values are estimates with associated margins of error per ACS methodology – consult source tables for confidence intervals. These data are presented for general context only and must not be used as a substitute for a formal market study or demographic analysis. Source: 2019–2023 ACS 5-Year Estimates.

Infrastructure & Access

Broadband Availability – FCC National Broadband Map

Required disclosure: FCC broadband data is self-reported by internet service providers and may significantly overstate actual service coverage and speeds at any given location. Treat these values as indicative only. Verify broadband availability and tier directly with each listed service provider before relying on this data for site selection, feasibility analysis, or development planning. Source: FCC National Broadband Map, deployment data.

Provider	Technology	Max Down (Mbps)	Max Up (Mbps)
AT&T	Fiber	5000	5000
Cox Communications	Cable	2000	100
Cox Communications	Fiber	2000	100
Verizon	Licensed Fixed Wireless	300	20
AT&T	Copper/DSL	100	20
AT&T	Licensed Fixed Wireless	100	20
Wake Wireless	LBR Fixed Wireless	100	20
Wake Wireless	Unlicensed Fixed Wireless	50	10
T-Mobile	Licensed Fixed Wireless	25	3
UNITED STATES CELLULAR CORPORATION	Licensed Fixed Wireless	25	3

Data Sources & Methodology

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Data Source	Version / Date	Refresh Cadence	Storage
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About This Report

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